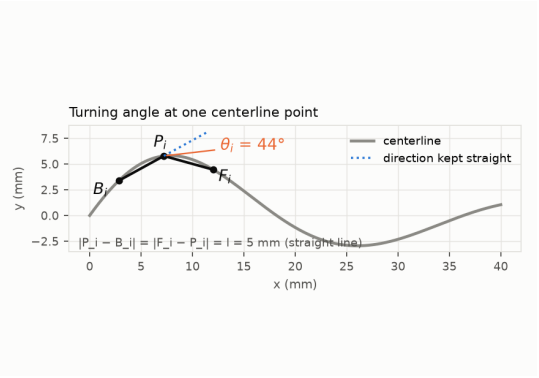
Curvature shapes in the proximal and mid LAD (FPCA)

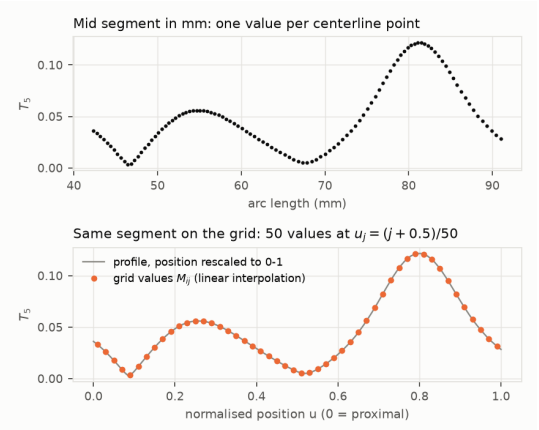
In the main model, curvature is the mean turning of each segment. The mean does not say where the vessel bends: one sharp bend and a gentle turn along the whole segment can give the same mean. Functional principal component analysis (FPCA) finds the few main shapes in which people's turning differs along a segment, and gives each person a score for how much of each shape they have. These scores then replace the mean in the risk model, to see whether where the vessel bends matters, not only how much. This is only done in the pLAD and mLAD, the only segments where enough people developed plaque to estimate several curvature terms, and it is descriptive. The figures use made-up people.

Turning angle. At each point of the centerline, one point is placed 5 mm before it and one 5 mm after it, in a straight line. The turning angle is the angle between the line coming in and the line going out: how much the vessel changes direction at that point. Dividing it by the 5 mm gives the turning in rad/mm. Bends shorter than about 5 mm are not seen, and the first and last 5 mm of a vessel get no value. It is computed along the whole artery and then cut at the segment boundaries, so a bend across a boundary is still measured.



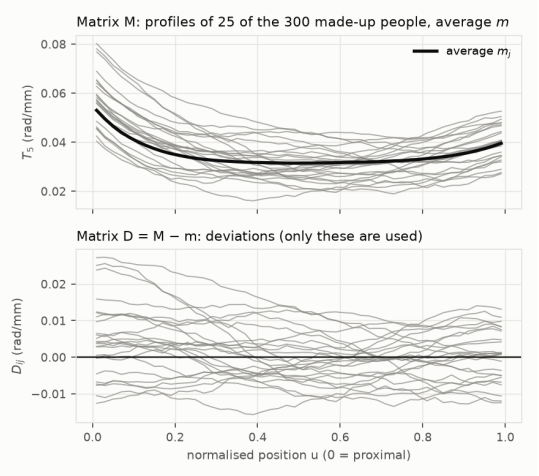
The turning angle θ_i at one point of a made-up centerline.

The same 50 places for everyone. The turning has one value per centerline point, and segments differ in length and in how many points they have (top). To compare people place by place, each segment is stretched or squeezed to run from 0 (start) to 1 (end), and the turning is read at the same 50 places (bottom). Place 25 is then halfway along the segment for everyone, whether the segment is 30 or 50 mm long. The shapes therefore describe where along the segment the vessel turns, not how long it is.



One made-up segment in mm (top) and read at the 50 places (bottom).

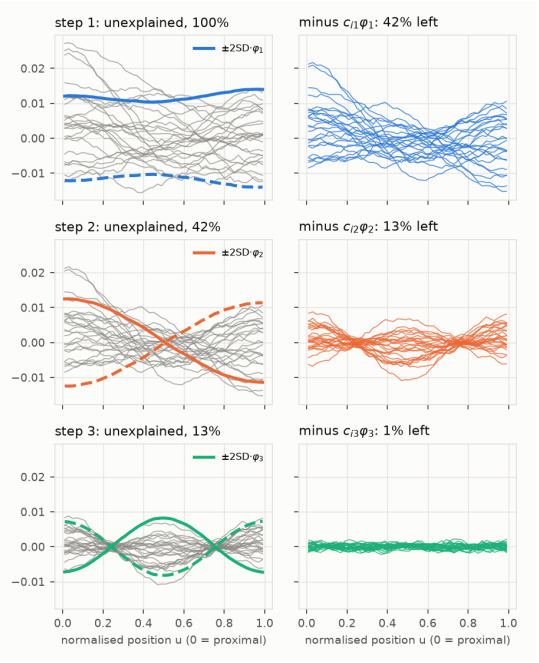
Turning curves and deviations. Each person is now one curve of 50 values (top, grey). The average curve (black) says how much everybody turns at each place, for example that everybody turns more near the start. It is the same for everyone, so it cannot tell people apart. At each place the average there is subtracted from everyone. What is left, the deviation (bottom), is how much and where each person turns more or less than the average person. Where the bundle is wide, people differ a lot. A bend covers several neighbouring places, so these places go up and down together across people; FPCA looks for such groups of places, and each group becomes a shape.



Turning curves of made-up people with their average (top), and the same curves minus the average (bottom).

Shapes, one at a time. The first shape is the pattern along which people differ most (top left). Each person has some amount of it, their score, and when that amount is taken away, less of the differences is left (top right). The next shape is found in what is left, and so on, so each shape describes something the earlier ones do not. Here the first shape is almost flat and positive, and the next shapes say *where* a person turns: more at the start than at the end, or more in the middle than at the ends.

Number of shapes. After the real shapes, new shapes only describe noise. Three checks decide where to stop: whether a shape explains more than it would by chance, whether it comes back when the people are redrawn, and how many shapes best predict a hidden part of a curve. The smallest of the three is used, and it is fixed before plaque is looked at, so the choice cannot be tuned to find an effect.



Left: what is not yet explained, with the next shape. Right: what is left.

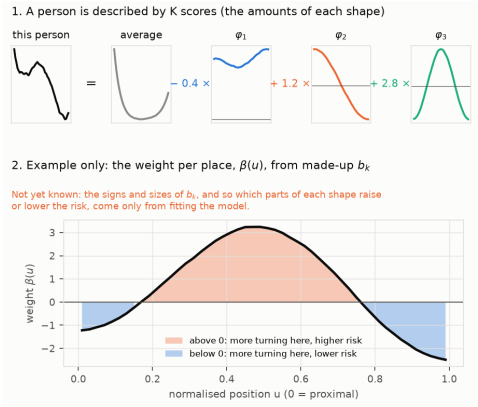
A flat first shape is the mean turning. When the first shape is a flat line, people mainly differ in that some turn more than average along the whole segment and others less. The score on this shape then only says how much more or less a person turns overall, which is the same as their mean turning. The first shape is only nearly flat, yet its score and the mean turning correlate 0.998 here.

We see the same in the Herlev–Østerbro data. In the Medis centerlines the first shape has the same sign along the whole segment and is close to a straight line, and explains the largest share of the differences, so the mean turning captures most of what separates them. The Medis trees come from participants with plaque at baseline, so this must be confirmed in the plaque-free cohort.

From scores to the risk model. The shapes are the same for everyone; people differ in how much of each they have. A score of 0 is the average person, +1 one SD more of the shape. The average plus each shape times the score rebuilds the curve (top), so a few scores replace the 50 numbers and go into the risk model of the segment, next to SCORE2:

$$\operatorname{cloglog}(p_i) = a + g\,\mathrm{SCORE2}_i + \sum_k b_k\, z_{ik},$$

where p_i is the risk that person i develops plaque in the segment and e^{b_k} the hazard ratio per SD of shape k. The weights read back as one weight per place (bottom). They are not estimated yet, so which parts of each shape raise or lower the risk is unknown.



Top: one made-up person as the average plus scores times shapes. Bottom: made-up weights; the real signs are not yet known.

The end goal. The aim is to see whether some patterns of turning carry more risk than others, that is, whether the weights of some shapes have more predictive value than others. A person who turns a lot at the start of the segment and hardly at all after it may carry more risk than a person who turns a little the whole way along, even with the same mean turning, and the mean turning cannot tell these two apart. The main model keeps the mean turning: it was chosen beforehand, as earlier studies point to it and it explains most of the differences between people, and it is not changed after this analysis, to avoid choosing the model by its p-values.